

Mruthyunjay Kubendran Sumathi

mruthyunjay [at] arizona.edu
University of Arizona
Dept. of Ecology and Evolutionary Biology
Tucson AZ 85705
ORCID: 0000-0003-3542-7816



EDUCATION

University of Arizona, Tucson, AZ	2021 - Present
PhD Candidate in Ecology and Evolutionary Biology	
University of Arizona, Tucson, AZ	2021 - Present
Minor in Astrobiology	
École Normale Supérieure, Université Paris Sciences & Lettres, Paris, France	
Interdisciplinary Masters (MS) in Life Science (magna cum laude)	
Concentration: Ecology and Evolutionary Biology	2019 - 2021
SRM Institute of Science and Technology, Chennai, TN, India	2014 – 2018
Bachelor (B.Tech) in Genetic Engineering (first class with distinction)	

FELLOWSHIPS, GRANTS & AWARDS

2026	GPSC Travel Grant, University of Arizona (\$700)
2026	University of Arizona Astrobiology Seed Grant “Bridging Biology and Astrobiology: A Perspective Article and AI-Assisted Literature Classification Tool” (\$3500)
2025	University of Arizona Galileo Circle Scholar (\$1000)
2024 – 2025	University of Arizona Carson Fellow (\$5000)
2024	University of Arizona Galileo Circle Scholar (\$1000)
2024	University of Arizona Astrobiology Seed Grant “Measuring the cost of cell maintenance to understand survival and adaptation” (\$9000)
2023	GPSC Travel Grant, University of Arizona (\$1000)
2022	GPSC Travel Grant, University of Arizona (\$1500)
2022	Université Paris-Saclay-CNRS internship (€1775)
2020	ENS-CNRS internship (€3600)
2020	EABISS Graduate Fellowship (€5000)
2019 – 2020	Qlife master Fellowship (€10000)

RESEARCH INTERESTS

Comparative Phylogenomics of Thermal Adaptation Across the Tree of Life

- Developing and scaling a phylogenetically corrected comparative genomics pipeline (PGLS, ancestral state reconstruction, evolutionary model fitting) across Bacteria and Archaea to test whether genome streamlining is an ancestral prerequisite for high-temperature life, or an ongoing adaptive response.

Experimental Approach to Metabolism Thermodynamics

- Anaerobic cultivation of the hyperthermophilic archaeon *Thermococcus kodakarensis*, growth-curve fitting, and ATP assays to estimate species-specific catabolic yield and maintenance energy cost across temperature.

Mathematical and Evolutionary Modeling of Adaptation to Extreme Environments

- **Bioenergetic modeling:** coupled ODE frameworks linking substrate uptake, ATP flux, and biomass production to temperature- and complexity-dependent maintenance cost, defining the thermodynamic boundaries of High- and Low-Energy Metabolizer strategies.
- **Adaptive dynamics:** evolutionary invasion analysis testing whether genome and proteome streamlining constitutes an Evolutionarily Stable Strategy under extreme thermal stress.

Microbial Syntrophy and the Evolution of Metabolic Interdependency

- Investigating how syntrophic partnerships evolve, including whether they can arise from a single ancestral lineage diverging into interdependent metabolic roles, and how thermodynamic constraints, together with quantum kinetic effects on enzyme catalysis at thermal extremes, shape that transition.

Proteome Allocation Modeling and the Thermal Limits of Photosynthesis

- Trait-based (Yield-Acquisition-Stress) and proteome-allocation modeling of why oxygenic and anoxygenic photosynthesis hit a functional ceiling near 75°C, incorporating Rubisco kinetic trade-offs and phylogenetic path analysis, supported by custom Python simulation and visualization tools.

Astrobiology: From Metabolic Constraints to Biosignatures

- Applying thermodynamic and genomic thresholds of viability to define expected biosignatures and detection limits for exoplanets, relative to instrument sensitivity thresholds such as the Habitable Worlds Observatory's 0.1 ppb contrast limit.

SKILLS

- **Bioinformatics & Comparative Genomics:** phylogenetic comparative methods (PGLS, OUwie, phylopath, ancestral state reconstruction) in R (*ape*, *geiger*, and related packages); large-scale genomic data curation and extraction from NCBI, BioCyc, and GTDB.
- **Programming & Computational Modeling:** Python and R for growth-curve model fitting, energetic parameter estimation, ODE-based bioenergetic simulation, and adaptive dynamics/evolutionary invasion analysis; data pipeline development and visualization (pandas, matplotlib, seaborn).
- **Statistical & Phylogenetic Analysis:** phylogenetic generalized least squares (Pagel's λ), Brownian motion and Ornstein-Uhlenbeck model fitting, phylogenetic path analysis, population dynamics modeling.
- **Microbial Culturing:** anaerobic culturing and handling of extremophilic archaea (*Thermococcus kodakarensis*), including dedicated anaerobic-chamber technique. Culturing and handling of marine and clinical microbes.
- **Molecular Techniques:** genome sequencing and Polymerase Chain Reaction (PCR); recombinant DNA technology; biomolecule extraction, protein purification, and electrophoresis; HPLC; ATP assays; microscopy (optical, fluorescence, confocal).
- **Other:** Git/GitHub version control; scientific writing and grant proposal development; science communication and curriculum design for K-12 and public audiences.

Languages: Sourashtra (native), Tamil (native), Hindi (fluent), English (fluent)

PUBLICATIONS

Bruneaux, M., Ashrafi, R., Kronholm, I., Laanto, E., Örmälä-Tiznado, A.-M., Galarza, J. A., Zihan, C., Kubendran Sumathi, M., & Ketola, T. (2022). The effect of a temperature-sensitive prophage on the evolution of virulence in an opportunistic pathogen bacterial. *Molecular Ecology*, 31, 5402– 5418. <https://doi.org/10.1111/mec.16638>

Kubendran Sumathi, M. (2018) Identifying potential genes in *S. marcescens* affecting virulence by gene prioritization and allelic replacement mutagenesis. B. Tech. Thesis, SRM Institute of Science and Technology, Chennai, TN, India.

Kubendran Sumathi, M. (2021) Towards a mechanistic eco-evolutionary modeling framework to understand the thermal limit of life. M. Sc. Thesis,

Manuscripts in Preparation

Kubendran Sumathi, M., Duhamel, S., Ferriere, R. Genome complexity and thermal niche co-evolution in Archaea.

Kubendran Sumathi, M., Gorlas, A., Guyot, F., Ferriere, R. Metabolic-dependent growth dynamics and bioenergetics of *Thermococcus kodakarensis*.

PROFESSIONAL PRESENTATIONS

External:

Kubendran Sumathi, M., Duhamel, S., Ferriere, R., 2026 Genome Reduction, Phylogeny, and the Evolution of High-Temperature Adaptation. Astrobiology Science Conference (AbSciCon), University of Wisconsin Madison, Madison, WI. *Oral Presentation*

Kubendran Sumathi, M., Duhamel, S., Ferriere, R., 2026 Quantifying the thermodynamic threshold of photosynthetic life via trait-based energy models. NASA NEXSS Extraterrestrial photosynthesis Workshop, University of Colorado, Boulder, CO. *Oral Presentation*

Kubendran Sumathi, M., Duhamel, S., Ferriere, R., 2025 Evolution of complexity across the entire life temperature gradient. Astrobiology Graduate Conference (AbGradCon), University of Colorado, Boulder, CO. *Oral Presentation*

Kubendran Sumathi, M., 2025, What is Astrobiology? Arizona Astrobiology Center (AABC), University of Arizona, Tucson, AZ. *Invited Talk*

Kubendran Sumathi, M., 2024, Unveiling Thermal Adaptation in Organisms: Insights from an Energy-Centric Perspective in Astrobiology. Astrobiology Graduate Conference (AbGradCon), Cornell University, Ithaca, NY. *Oral Presentation*

Kubendran Sumathi, M., Guyot, F., Duhamel, S., Ferriere, R., 2024, Exploring High-Temperature Adaptation through the 'Follow the Energy' Approach for Astrobiology. Astrobiology Science Conference (AbSciCon), Providence, RI. *Oral Presentation*

Kubendran Sumathi, M., Guyot, F., Duhamel, S., Ferriere, R., 2024, Investigating Thermal Adaptation in Organisms: Insights from Astrobiology's Energy-Centric Approach. 4th Annual Arizona Astrobiology Symposium, ASU, Tempe AZ. *Poster Presentation*

Kubendran Sumathi, M., Gorlas, A., Guyot, F., Ferriere, R., 2022, Metabolic dependent growth dynamics of *T. kodakarensis*. International Congress on Extremophiles, Loutraki, Greece. *Oral Presentation*

Internal (Home Institution):

Kubendran Sumathi, M., Duhamel, S., Ferriere, R., 2025 A Genome-First Model for the Adaptation of Archaea to High Temperatures. University of Arizona Ecology and Evolutionary Biology Dept. Open House, Tucson AZ. *Poster Presentation*

Kubendran Sumathi, M., 2025, Temperature-dependent energy trade-off in Microbes: Patterns in metabolic complexity across temperature gradients. University of Arizona Ecology and Evolutionary Biology Dept. Seminar, Tucson AZ. *Oral Presentation*

Kubendran Sumathi, M., 2024, Understanding High Temperature Adaptation: Insights from patterns of cellular complexity. University of Arizona Ecology and Evolutionary Biology Dept. Seminar, Tucson AZ. *Oral Presentation*

Kubendran Sumathi, M., 2023, Microbial Life at thermodynamic limits: Low energy metabolism and adaptation. University of Arizona Ecology and Evolutionary Biology Dept. Seminar, Tucson AZ. *Oral Presentation*

Kubendran Sumathi, M., 2022, Is there a thermal limit to life adaptation? University of Arizona Ecology and Evolutionary Biology Dept. Seminar, Tucson AZ. *Oral Presentation*

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2020	EABISS Graduate Fellowship (€5000)
2019 – 2020	Qlife master Fellowship (€10000)

TEACHING

2026	Teaching Assistant, <i>Evolution</i> , University of Arizona
2025	Teaching Assistant, <i>Biology of the Oceans</i> , University of Arizona
2025	Teaching Assistant, <i>Evolution</i> , University of Arizona
2024	Teaching Assistant, <i>Genetics</i> , University of Arizona
2024	Teaching Assistant, <i>Evolution</i> , University of Arizona
2022	Teaching Assistant, <i>Geneics</i> , University of Arizona
2022	Teaching Assistant, <i>Evolution</i> , University of Arizona
2021	Teaching Assistant, <i>Introductory Biology Laboratory</i> , University of Arizona

SERVICE & OUTREACH

2025 – present	Sky School Instructor
2025 – present	Astronomy Curriculum Development for high school students
2025 – present	Astrobiology Curriculum Development for Juvenile detention center
2025	Co-Organiser, Viking 2.0 Mock Data Workshop
2025	Co-Organiser, NASA Astrobiology Graduate Conference
2024 – 2025	Finance Chair, University of Arizona, EEB Graduate Council
2022 – 2024	Resource Personel, Meet MCB program for High School Students

2022 – present	Co-organizer, University of Arizona Astrobiology Journal Club
2021 – present	Participating Scientist, SkypeAScientist Program
2014 – 2017	Instructor, Hands On Genetics Workshop for High School Students
2016	Ambassador, Green (R)evolution Climate Change Awareness Program

ADDITIONAL TRAININGS & LANGUAGES

- 2022 Culturing and handling of anaerobe, *T. kodakarensis* at CNRS - Paris-Saclay University
- 2017 Handling and experimenting with mice at Kavins Bioresearch, India
- 2016 Culturing & Handling of Marine Microbes at Apex Bioresearch Institute, India
- 2015 Molecular Diagnosis, Counseling, and Research Training at MDCRC, India

Languages: Sourashtra (native), Tamil (native), Hindi (fluent), English (fluent), French (beginner A1/2)

SUPPORTED PROJECTS

Eco-evolutionary drivers of diversity in toxic algal blooms (Research Assistant, University of Arizona, Tucson, USA)

PI: *Dr Regis Ferriere*

Towards a mechanistic eco-evolutionary modeling framework to understand the thermal limit of life. (MS Thesis - École Normale Supérieure, Université PSL, Paris, France)

PI: *Dr Regis Ferriere* Supervisor: *Dr François Guyot*

Identifying potential genes in *S. marcescens* affecting virulence by gene prioritization and allelic replacement mutagenesis. (B.Tech. Thesis - University of Jyväskylä, Jyväskylä, Finland)

PI: *Dr Tarmo Ketola* Supervisor: *Dr Ilkka Kronholm*

Expression Study of GH5 family genes from *C. phytofermentans* in *E. coli* BL21 DE3 and characterizing its cellulolytic ability. (B.Tech. Minor Project, SRMIST, Chennai, TN, India)

PI and Supervisor: *Dr Ramya Mohandass*

STUDENTS MENTORED

Gisselle Ramirez (Undergraduate, University of Arizona)